

4.4. MEASUREMENT OF THE TRANSVERSE BEAM PROFILE

Often it is desired to expand or reduce the diameter of a laser beam while maintaining the parallelism, the *collimation*, of the beam. This can be achieved with a pair of lenses arranged as a "telescope." Telescopes were invented by Galileo and by Newton who used them to achieve *angular* magnification; the same arrangements are still in use and are named after their discoverers. To calculate the magnification of the beam we will use only geometrical optics; this is sufficient for our present considerations, even though a Gaussian laser beam diverges due to diffraction effects (see Fig. 4.6).

Figure 4.11 shows the *Newtonian* (or astronomical) telescope consisting of two focusing lenses of focal length f_1 and f_2 . As shown in the sketch the lenses are converging (plano-convex), and the distance between them is

$$\ell = f_1 + f_2. \quad (4.25)$$

If a collimated beam of diameter d_1 is incident from the left, parallel to the telescope axis, it will exit with a diameter d_2 , where

$$d_2 = d_1 \frac{f_2}{f_1}. \quad (4.26)$$

By appropriate choice of f_1 , f_2 we can magnify or demagnify the beam.

The *Galilean* telescope is shown in Fig. 4.12; a diverging (plano-concave) lens of focal length f_1 and a converging lens of focal length f_2 are used. To preserve collimation the distance between the lenses must be

$$\ell = f_2 - f_1. \quad (4.27)$$

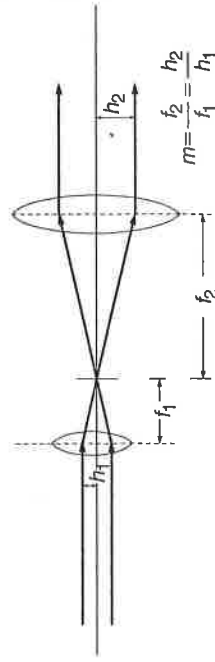


FIGURE 4.11 A Newtonian telescope with magnification f_2/f_1 .

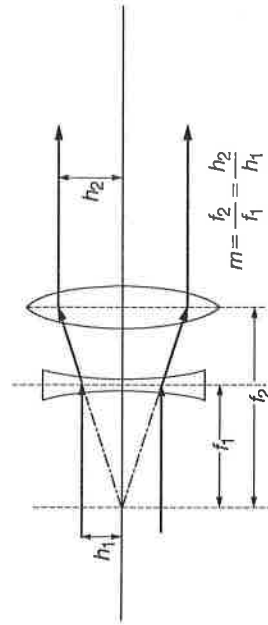


FIGURE 4.12 A Galilean telescope with magnification f_2/f_1 .

and the spatial magnification is given by

$$d_2 = d_1 \frac{f_2}{f_1}. \quad (4.28)$$

The curved surface of the lens, whether convex or concave, is spherical, and the focal length is related to the radius of curvature, R , through the lens-makers equation. When the second surface is plane,

$$\frac{1}{f} = \frac{n-1}{R}, \quad (4.29)$$

where n is the index of refraction of the lens material. For most glasses used in lens manufacture and for visible light we can approximate $n \simeq 1.5$, so that $f \sim 2R$.

In setting up a telescope certain "alignment tricks" are useful. The beam must pass through the center of both lenses. Thus the lenses must be set on the optical table at the same height as the laser. In the horizontal direction one can be helped by noting that a beam that is passing through a lens off-center is steered. Furthermore, the surface of the lens must be perpendicular to the beam axis; this is most easily achieved by back-reflecting the beam.

The transmitted intensity of the beam is measured by a photodiode. (See Appendix E.) Since the photodiode area is small, it is often necessary to focus the beam on it, especially if it has been expanded. The diode is backward biased, usually with a low-voltage battery as shown in Fig. 4.13. With no incident light R_D is infinite. When light is incident some carriers are liberated and the resistance R_D of the diode decreases. Therefore the voltage across the load varies as

$$V_{\text{out}} = V_B \frac{R_L}{R_L + R_D}. \quad (4.30)$$

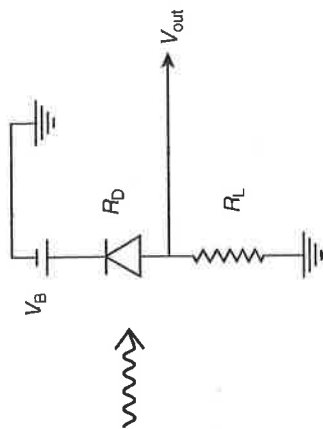
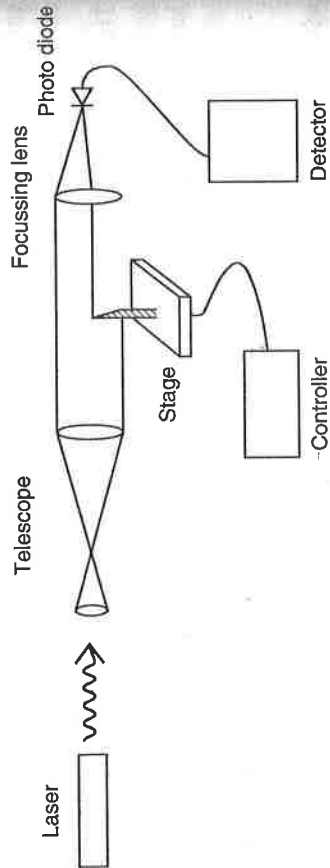
FIGURE 4.13 A photodiode reverse biased by a source V_B and working into a load R_L .

FIGURE 4.14 Arrangement for measuring the transverse profile of a laser beam.

At low light levels (i.e., where R_D is large) a digital voltmeter ($R_L \simeq 10\text{ M}\Omega$) is adequate to read V_{out} . At high light levels the diode may become saturated, and it is desirable to use a shunt resistor; the signal can also be viewed on an oscilloscope, but when fast response is desired, as in Fig. 4.10, a 50- Ω impedance must be maintained throughout. When working at low light levels the photodiode must be shielded against room light.

One way of measuring the beam profile is to record the intensity received at the photodiode as a sharp edge (i.e., a razor blade) is moved through the beam. The blade is mounted on a translation stage that can be positioned with a resolution of few micrometers. The arrangement is sketched in Fig. 4.14, and the recorded intensity as a function of position gives the integral of the beam profile. If the beam profile in the direction of the blade motion is

$$I(x) = I_0 g(x) \quad (4.31)$$

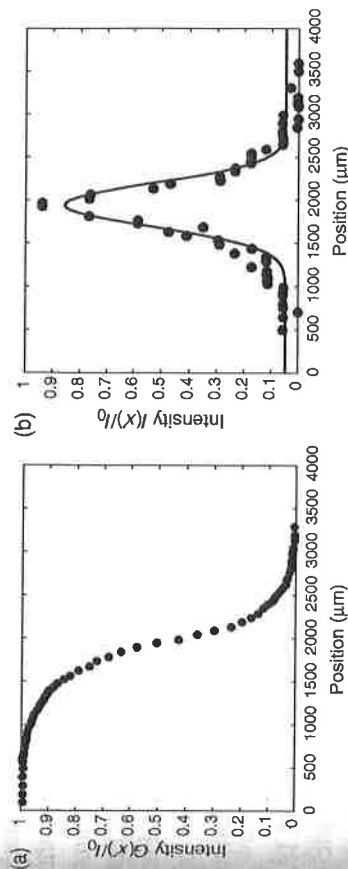


FIGURE 4.15 (a) The transmitted intensity as a function of the position of the obstacle (razor blade), which is moved across the beam. (b) The derivative of (a) gives the transverse profile of the beam intensity. It is fitted by a Gaussian.

with $\int_{-\infty}^{\infty} g(x) dx = 1$, the transmitted intensity when the blade is at position x' is

$$G(x') = I_0 \int_{x'}^{\infty} g(x) dx \quad (4.32)$$

(when the beam is fully unmasked, $x' \rightarrow -\infty$).

A typical result for the laser beam is shown in Fig. 4.15a where measurements were taken every 100 μm . By differentiating $G(x')$ we recover the intensity profile

$$-\frac{d}{dx'} G(x') = I(x'). \quad (4.33)$$

Performing this operation on the data of Fig. 4.15a we obtain the result shown in Fig. 4.15b, which can be adequately fitted by a Gaussian. The $1/e^2$ points of the Gaussian define the beam diameter, which in this case is $2w = 1000\text{ }\mu\text{m}$.

4.5. THE MICHELSON INTERFEROMETER

We are familiar with the fact that wave phenomena exhibit interference; namely at every point in space the amplitudes of two waves are added linearly (they are superimposed), whereas the intensity is determined